



RCamp: Introductory Applied Statistics

Tribal-State Relations Acknowledgment Statement

The State of Minnesota is home to 11 federally recognized Indian Tribes with elected Tribal government officials. The State of Minnesota acknowledges and supports the unique political status of Tribal Nations across Minnesota and their absolute right to existence, self-governance, and self-determination. This unique relationship with federally recognized Indian Tribes is cemented by the Constitution of the United States, treaties, statutes, case law, and agreements. The State of Minnesota and Tribal governments across Minnesota significantly benefit from working together, learning from one another, and partnering where possible.

Minnesota Department of Health recognizes, values, and celebrates the vibrant and unique relationships between the 11 Tribal Nations and the State of Minnesota. Partnerships formed through government-to-government relationships with these Tribes will effectively address health disparities and lead to better health outcomes for all of Minnesota.

In our work at the Office of Data Strategy and Interoperability, we demonstrate our commitment to Tribal-State relations by providing free assistance upon request and promoting health equity in data collection and use.

Purpose of DSI

To provide vision, direction, and leadership in advancing data strategies and data exchange across MDH through:

- Coordinating and streamlining the exchange of data with MDH
- Overseeing and supporting state solutions and common tools
- Facilitating efforts to maximize MDH data by creating data and process standards and tools with the whole Minnesota Public Health System in mind

The purpose of DSI is to provide vision, direction, and leadership across MDH. We accomplish this by streamlining data exchange between MDH and outside partners, including LPH, advocating for common tools and standards within and with MDH, and working toward solutions that help LPH access and use MDH data for their work.

Vision of DTA

To support **data needs** for ALL staff in:

- MDH
- Local Public Health
- Tribal Health

Data needs we support:

- Project planning and design
- Data wrangling and analysis
- Data visualization and report preparation
- Data literacy

10/6/2026

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4

The vision of our unit is to support a range of data needs for ALL government staff working in public health in Minnesota. If you are familiar with the data lifecycle, we referenced that when determining what we could support. We support project design and conceptualization, data cleaning and validation, data analysis, and data visualization. We also provide general support through venues like trainings and communities of practice.

Thank you to the State of Minnesota R user group!

★ Special thanks to our founders Barbara Monaco, Dorian Kvale, Derek Nagel, and Kristi Ellickson.

Introductory Applied Statistics Overview

- Today we will cover:

- Types of data, summary statistics, and distributions
- Hypothesis testing and P-value
- T-tests, Mann-Whitney U Test, Wilcoxon Signed Rank Test
- ANOVA
- Chi-Squared Test
- Correlation
- Linear Regression



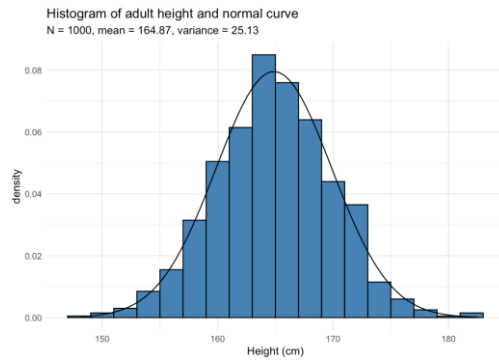
Types of Quantitative Data

- Continuous data
 - Data that can take on any value within a given range, including decimals and fractions (height, weight, temperature)
 - Might or might not be normally distributed
 - We will mostly focus on continuous data today
- Discrete data
 - Data that can only take on specific, separate values, typically whole numbers or categories
 - Binary data: discrete data with only two possible outcomes (e.g. dead or alive, sick or healthy)
 - Counts: whole numbers representing the number of events, patients, hospitals, or anything else that can be counted
 - Nominal data: data categories with no order (blood type, treatment group)
 - Ordinal data: data categories with a particular order (pain scale, survey response categories)

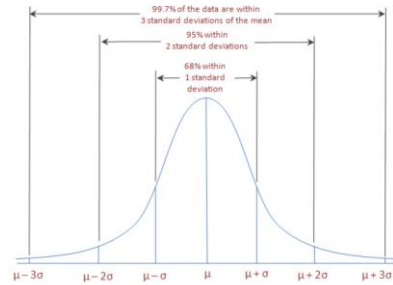
Summary Statistics

- Summary statistics summarize and provide information about your sample data.
- Summary statistics fall into three main categories:
 - Measures of **location**: where your data is centered at, or where a trend lies
 - Mean, median, mode
 - Measures of **spread** or distribution shape: how spread out or varied your data set is
 - range, standard deviation, skew and kurtosis
 - Graphs/charts: ways to display summary data
 - histogram, frequency distribution table, box plot, bar chart, scatter plot and pie chart

Normal Distribution



- $\mu \pm \sigma$ includes approximately 68% of the observations
- $\mu \pm 2 \cdot \sigma$ includes approximately 95% of the observations
- $\mu \pm 3 \cdot \sigma$ includes almost all of the observations (99.7% to be more precise)



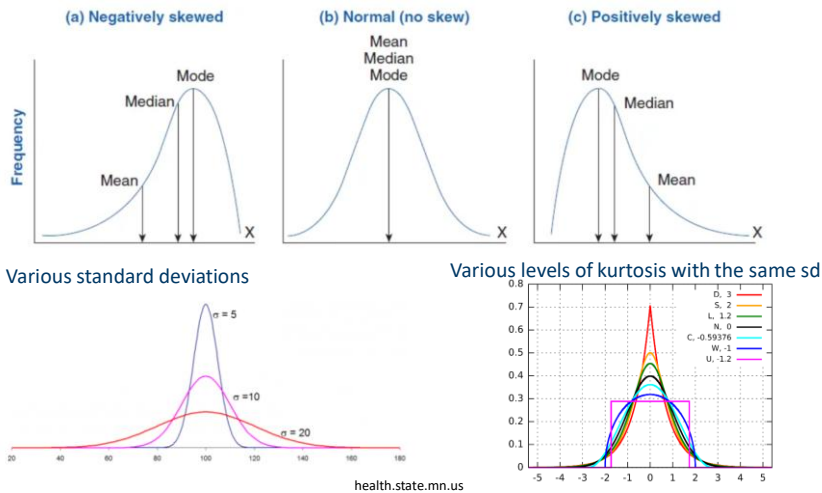
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<https://statsandr.com/blog/do-my-data-follow-a-normal-distribution-a-note-on-the-most-widely-used-distribution-and-how-to-test-for-normality-in-r/>

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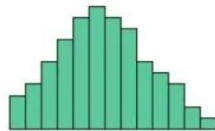
9

Skew, Standard Deviation, and Kurtosis

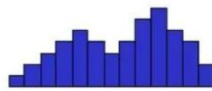


These things are important to keep in mind when exploring your data because whether a certain test is appropriate to use or not often depends on whether Your data is normally distributed or not

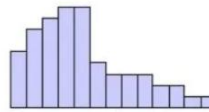
Non-Normal Distribution



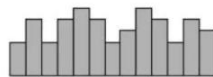
The shape has a bell shape.
It is **symmetric**.



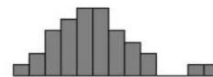
The shape has two humps.
It is **bimodal**.



The shape has a long tail.
It is **not symmetric**.



The shape is **flat**.



There are one or more **outliers**.

<https://leanscape.io/data-distributions-explained-what-are-the-different-types-of-distribution/>

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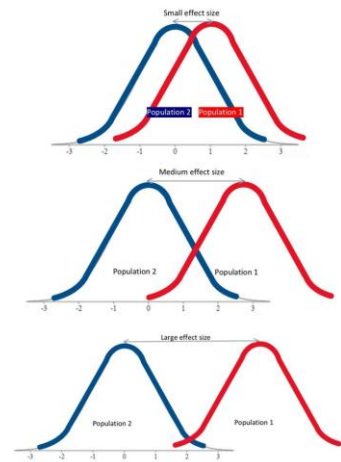
11

For many statistical tests, normal distribution is required in order to get accurate results!

Hypothesis Testing and P-value

- For example, we want to know if the average height between two groups of people is different or not
 - Null Hypothesis (H_0): There is no difference in average height between the two groups
 - Alternative Hypothesis (H_A or H_1): There is a difference in average height between the two groups
- P-value (probability value) tells us how likely is the difference in height that we see if the null hypothesis is true (there is actually no difference)
 - A commonly accepted P-value for statistical significance is equal to or below $\alpha = 0.05$
 - That means that 5% of the time the height difference that you see would be there by random chance even if the null hypothesis was true (error rate you are willing to accept)
 - You are 95% confident that there is a statistical difference in height between the groups

10/6/2026



12

P-value: probability of observing data as extreme (or more) than observed if H_0 were true.

Significance level (α): a pre-set threshold (commonly 0.05) for declaring statistical significance.

Error Types and Statistical Power

- Type I error: incorrectly reject H_0 when it is true (false positive)

- α is the probability of Type I error

- Type II error: fail to reject H_0 when it is false (false negative)

- β is the probability of Type II error

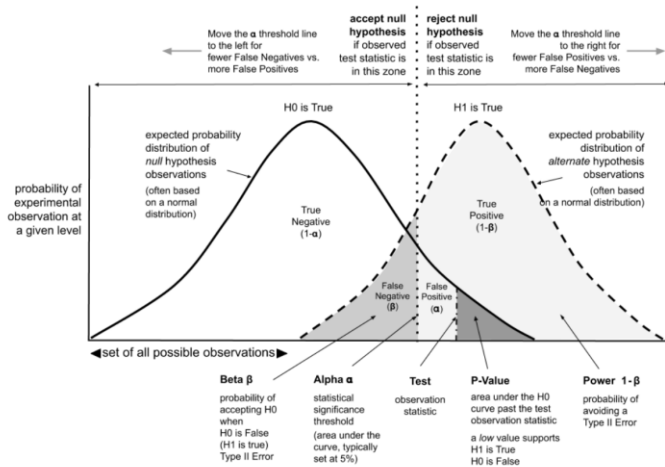
- Power = $1 - \beta$

- Probability that a test will correctly reject the H_0

- Visualization to explore:

- <https://rpsychologist.com/d3/nhst/>

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<https://www.ml-science.com/statistical-power-of-a-test>

P-value: probability of observing data as extreme (or more) than observed if H_0 were true.

Significance level (α): a pre-set threshold (commonly 0.05) for declaring statistical significance. It's just the probability of Type I error we are willing to accept.

T-Test

- Compares the means of two groups and tells whether the difference is significant
- Lets you know the likelihood those differences could have happened by random chance (P-value)
- Different kinds of t-tests:
 - A two tailed t-test checks whether there is a general difference between two groups.
 - A one tailed t-test checks whether one group is specifically larger or smaller (has a direction)

One sample t-Test



Is there a **difference** between a **group** and the **population**

Independent samples t-Test



Is there a **difference** between **two groups**

Paired samples t-Test



Is there a **difference** in a **group** between **two points in time**

T-tests were first developed to test the consistency of Guinness beer!

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<https://datatab.net/tutorial/one-sample-t-test>

14

A paired t-test assumes that there is less variability between the two measured groups, since it assumes that we are measuring the same subjects twice. By removing this variability that would usually exist if the groups were independent, we make the test more sensitive, and it often results in a lower p-value.

T-Test Assumptions

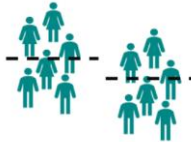
- Assumptions are extremely important in statistics! Tests are only valid when **all** the assumptions are met.
 - The data is numeric and continuous
 - The data comes from a random, representative sample
 - The data is normally distributed
 - To formally test for normality, you can use the Shapiro-Wilks test. A p-value < 0.05 suggests that the data is not normally distributed
 - The variance of data in both groups is approximately equal
 - The data in two groups is independent from each other (only relevant for the non-paired tests)
- What if we can't meet some of those assumptions?

Mann-Whitney U-Test

- If the data isn't normally distributed, you can use the Mann-Whitney U-Test
- The independent t-test counterpart for data that is not normally distributed.
- Tests that require assumptions about data distributions are called **parametric**. Tests that do not are called **nonparametric**.
- Instead of comparing groups directly, the test ranks everyone in both groups from shortest to tallest, sums the ranks in both groups, and compares this rank sum between the groups.

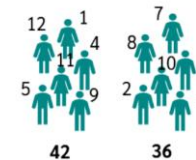
t-Test

Is there a difference in mean?



Mann-Whitney U Test

Is there a difference in the rank sum?



Wilcoxon Signed Rank Test

- What if the data is not normally distributed but is paired?
 - Use the Wilcoxon Signed Rank Test, the nonparametric counterpart to the paired t-test
 - Works just like the Mann-Whitney u-Test by comparing rank sums
- There are more types of tests for different situations, but these are some of the most common when comparing **two** groups or **one** sample group and the general population.
- What if you want to compare more than two groups at a time?

ANOVA

- Analysis of Variance (ANOVA) can be used to compare the means of more than two groups at a time

- One-way ANOVA compares **one** variable across multiple categories

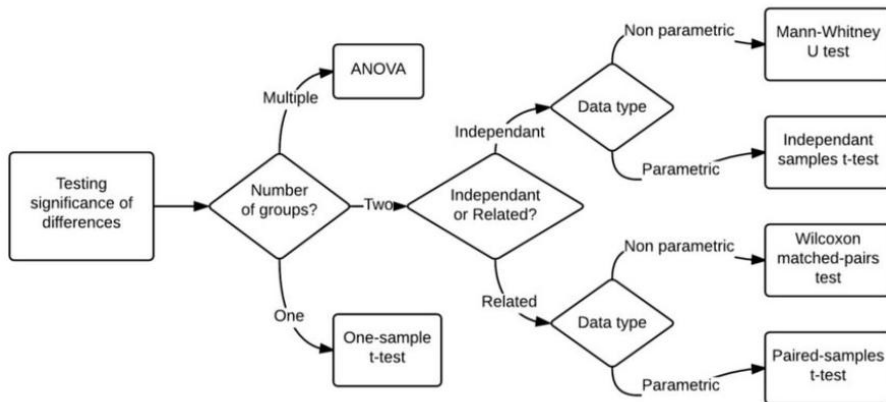
Example: Average disease duration in three cohorts receiving different treatment

	Men	Women
Cohort 1	9 days	10.2 days
Cohort 2	7.2 days	7.0 days
Cohort 3	5.6 days	4.8 days

- Two-way ANOVA compares **two** variables across multiple categories, as well as their interaction

Example: Average disease duration in three cohorts receiving different treatment and accounting for patient sex

Statistical Test Decision Tree



Chaudy, Yaelle. (2015). An Assessment and Learning Analytics Engine for Games-based Learning. 10.13140/RG.2.1.4932.5040.

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19

Chi-Square Test

- If your data isn't continuous but is instead categorical, you can use the Chi-Square test
 - Example: we want to tell whether out of a sample of 300 people, diabetes rates are different between men and women
 - H_0 : Diabetes rate are the same
 - H_A : Diabetes rate are different
- Assumptions:
 - Both datasets are categorial
 - Observations are independent
 - Each category (cell in the table) is mutually exclusive
 - Each value in the cell should be 5 or greater

	Diabetes	No Diabetes	Total
Men	43	114	157
Women	34	109	143
Total	77	223	300

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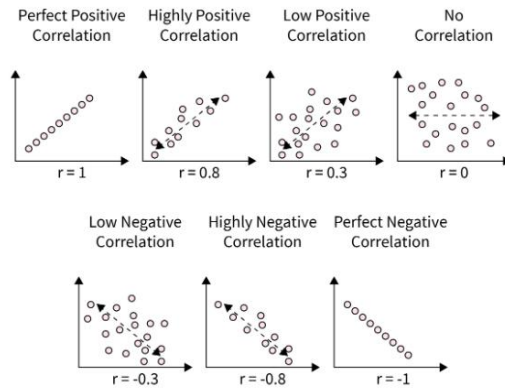
20

Make a contingency table

- Correlation describes the linear relationships between quantitative variables.
- **Pearson** correlation coefficient formulas (often shown as " r ") are used to find the strength of a relationship between two normally distributed continuous variables. The formula returns a value between -1 and 1, where:
 - 1 indicates a strong positive relationship.
 - -1 indicates a strong negative relationship.
 - A result of zero indicates no relationship at all.
- **Spearman** correlation is used for data that is not normally distributed, is non-linear, or has significant outliers. Like other non-parametric alternative tests, it is based on rank rather than raw values

Correlation Visual Examples

Scatter Plots & Correlation Examples



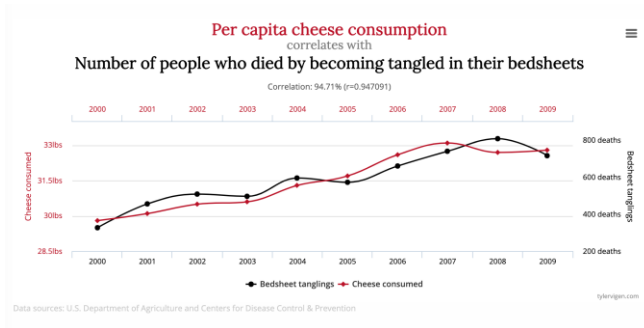
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<https://www.scaler.com/topics/data-science/covariance-and-correlation/>

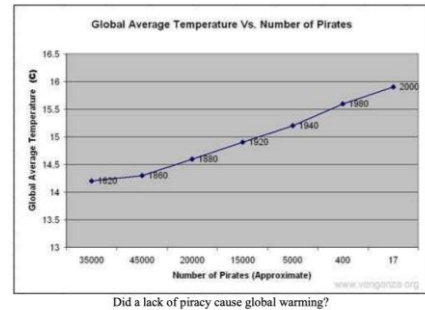
22

Correlation is not Causation

- Correlation is not causation! Just because two things correlate does not necessarily mean that one causes the other.



10/6/2026



23

Palmer Penguin Dataset

- Now let's look how these kinds of tests can be run in R
- This data was collected between 2007 and 2009 on the Palmer archipelago in the Antarctic
- <https://journal.r-project.org/articles/RJ-2022-020/>



species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex	diet	life_stage	health_metrics	year
Adelie	Biscoe	53.4	17.8	219	5687	female	fish	adult	overweight	2021
Adelie	Biscoe	49.3	18.1	245	6811	female	fish	adult	overweight	2021
Adelie	Biscoe	55.7	16.6	226	5388	female	fish	adult	overweight	2021
Adelie	Biscoe	38	15.6	221	6262	female	fish	adult	overweight	2021
Adelie	Biscoe	60.7	17.9	177	4811	female	fish	juvenile	overweight	2021
Adelie	Biscoe	35.7	16.8	194	5266	female	fish	juvenile	overweight	2021
Adelie	Biscoe	61	20.8	211	5961	female	fish	adult	overweight	2021
Adelie	Biscoe	66.1	20.8	246	6653	male	fish	adult	overweight	2021
Adelie	Biscoe	61.4	19.9	270	6722	male	fish	adult	overweight	2021

Regression Analysis

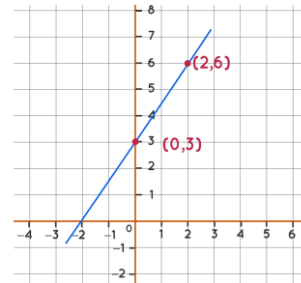
- Regression analysis provides you with an equation for a graph so that you can make predictions or draw insight about your data.
- Simple regression analysis uses a single x variable for each dependent “y” variable.
- Multiple regression analysis is used to see if there is a statistically significant relationship between sets of variables. It’s used to find trends in those sets of data.
- Regression is often more powerful and flexible than many other statistical tests and is a staple of statistical analysis

Linear Regression

- Linear regression equation:

$$y = b_0 + b_1 * x + \epsilon, \epsilon \sim \text{Norm}(0, \sigma^2)$$

- y – dependent variable
- x – independent variable
- b_1 – slope
- b_0 – intercept
- ϵ – the error term that represents all the variation in y that the model cannot explain



- Linear regression error term is normally distributed

$$\epsilon \sim \text{Norm}(0, \sigma^2)$$

- 0 – the mean of the normal distribution
- σ^2 – standard deviation squared, also known as variance

- Linear regression visualization: https://ryansafner.shinyapps.io/ols_estimation_by_min_sse/

Linear Regression Assumptions

- The dependent variable (outcome) is continuous
- The relationship between predictors (x) and outcome (y) is linear
- The residuals of the regression (differences between observed and estimated values) are normally distributed
 - Note that the data itself does not have to be normally distributed
- No multicollinearity: independent variables should not correlate with each other more than ~ 0.8
- Homoscedasticity: the variance of the errors is constant across all levels of the independent variable
- Error terms are independent of each other

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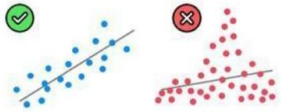
27

*Error dependence might be especially prevalent in time series data where earlier values influence future values

Linear Regression Assumptions

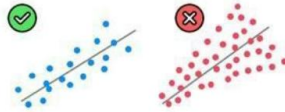
1. Linearity

(Linear relationship between Y and each X)



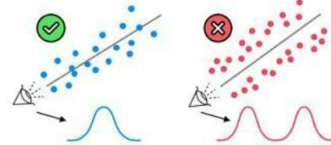
2. Homoscedasticity

(Equal variance)



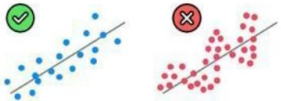
3. Multivariate Normality

(Normality of error distribution)



4. Independence

(of observations. Includes "no autocorrelation")



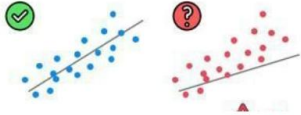
5. Lack of Multicollinearity

(Predictors are not correlated with each other)



6. The Outlier Check

(This is not an assumption, but an "extra")



<https://www.geeksforgeeks.org/machine-learning/assumptions-of-linear-regression/>

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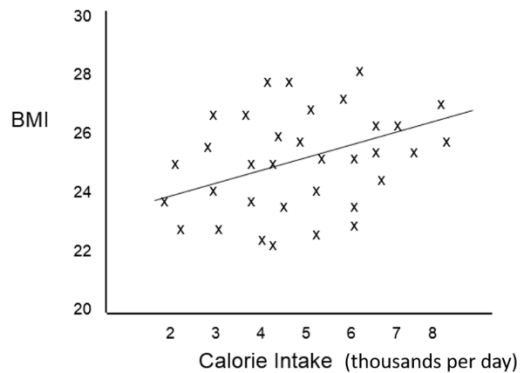
28

Multiple Linear Regression

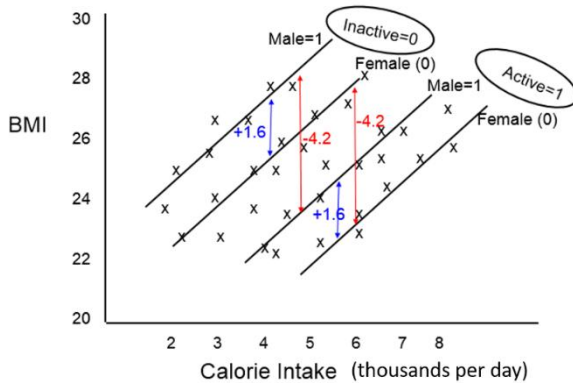
Consider a hypothetical situation evaluating BMI and Calorie Intake.

Multiple R squared value (a statistic to help assess the wellness of the regression fit) in this case was 0.12 , so caloric intake alone does not predict BMI well.

What other variables could influence BMI? Perhaps activity level and gender?



Multiple Linear Regression



The equation for this would be:

$$\text{BMI} = 15.0 + 1.5 (\text{cal}) + 1.6 (\text{if male}) - 4.2 (\text{if active})$$

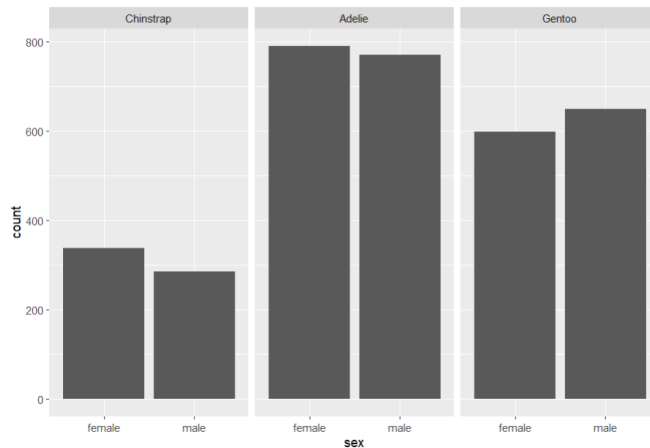
What can be said about active and inactive populations as it relates to gender, calorie intake, and BMI?

Solving this work by hand is very tedious, so we will use R

Many additional variables can be added into the equation to improve the model outcome

Exploring the Palmer Penguin Data

- Number of penguins by sex and species

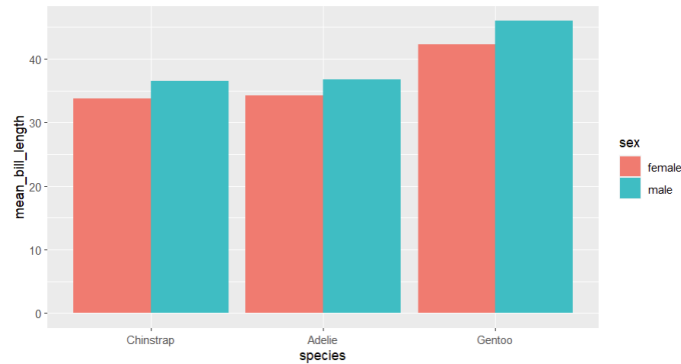


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31

It's usually the best practice to start exploring your data via figures, tables, and graphs and only move on to statistics once you understand it well enough

Exploring the Palmer Penguin Data

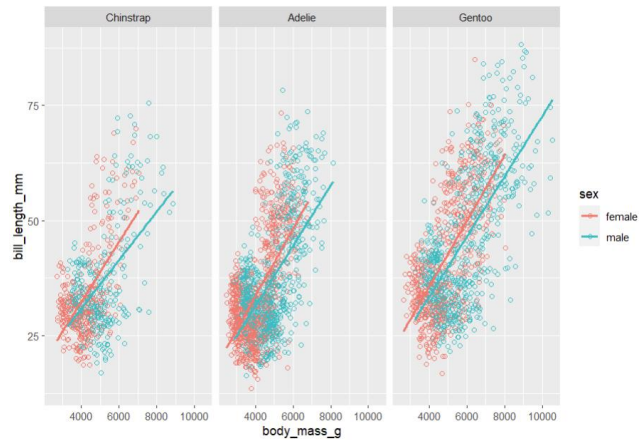


species	Chinstrap	Chinstrap	Adelle	Adelle	Gentoo	Gentoo
sex	female	male	female	male	female	male
mean_bill_length	33.8	36.5	34.2	36.7	42.2	46.0
mean_mass	4215.9	5061.0	4084.0	4816.4	4911.5	5922.5

A table can be more useful than a graph if you want to see specific numbers

Exploring the Palmer Penguin Data

The relationship between penguin bill length, body mass, species, and sex



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33

Graphs like these can help visualize complex relationships and build hypotheses. Interpretations? It seems like the relationship between body mass and bill length varies by sex and species

Linear Regression Interpretation 1

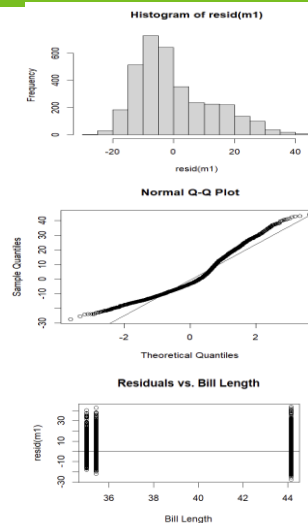
```
Call:
lm(formula = bill_length_mm ~ species, data = peng)

Residuals:
    Min       1Q   Median       3Q      Max
-27.362  -8.832  -3.215   7.538  44.038

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   35.4316     0.3157  112.214  <2e-16 ***
speciesChinstrap -0.4170     0.5911  -0.706    0.481
speciesGentoo    8.7303     0.4737  18.429  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 12.47 on 3427 degrees of freedom
Multiple R-squared:  0.1045,    Adjusted R-squared:  0.104
F-statistic: 200.1 on 2 and 3427 DF,  p-value: < 2.2e-16
```

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Interpretation: mean bill length of Adeline is 35.4mm. Gentoo bill length is 8.73mm longer than Adeline, which is statistically significant at $\alpha = 0.05$. The mean Chinstrap bill length is shorter than Adeline, but this difference isn't statistically significant.

R-squared: the proportion of the variance of the dependent variable that can be explained by the independent variable. It is 0.104, so only about 10% of the bill length is explained by the species of penguin. Multiple R-squared always increases with additional variables, so Adjusted R-squared is generally a better statistic to consider. Regression models should aim to be parsimonious.

QQ-plot and histogram both show that the residuals aren't quite normally distributed. QQ-plot plots the quantiles of the residuals against a theoretical distribution. It is best if it is a straight line.

Shapiro-Wilk test might be useful to formally test the normality of these residuals.

Residuals vs Bill Length is showing the bill length on the x-axis, and how residuals are dispersed for the tree species (three vertical lines). The variance of the residuals is

fairly constant, which is good.

Exploring the Palmer Penguin Data

Which species should serve as the reference category for the regressions?

```
# A tibble: 3 × 2
  species mean_bill_length
  <chr>      <dbl>
1 Adelie    35.4
2 Chinstrap 35.0
3 Gentoo    44.2
```

Linear Regression Interpretation 2

```
Call:
lm(formula = bill_length_mm ~ species, data = peng)

Residuals:
    Min       1Q   Median       3Q      Max
-27.362  -8.832  -3.215   7.538  44.038

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  35.0146    0.4996   70.079  <2e-16 ***
speciesAdelie  0.4170    0.5911    0.706   0.481
speciesGentoo  9.1473    0.6119   14.950  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 12.47 on 3427 degrees of freedom
Multiple R-squared:  0.1045,    Adjusted R-squared:  0.104
F-statistic: 200.1 on 2 and 3427 DF,  p-value: < 2.2e-16
```

10/6/2026

36

Average Chinstrap bill length is 35.01mm. The length of Gentoo beak is 9.15mm longer than Chinstrap, which is statistically significant at $\alpha = 0.05$. Adelie bill length isn't statistically significantly different from Chinstrap.

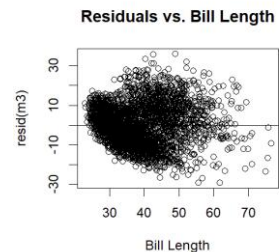
Linear Regression Interpretation 3

```
Call:
lm(formula = bill_length_mm ~ body_mass_g, data = peng)

Residuals:
    Min       1Q   Median       3Q      Max
-29.134  -7.317  -0.517   6.759  36.037

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  6.4074805   0.6450241    9.934  <2e-16 ***
body_mass_g  0.0066441   0.0001288   51.598  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.886 on 3428 degrees of freedom
Multiple R-squared:  0.4371,    Adjusted R-squared:  0.437
F-statistic: 2662 on 1 and 3428 DF,  p-value: < 2.2e-16
```



10/6/2026

37

What does the intercept 6.407 mean? Does this have scientific significance?

- It is the average length of the bill of all the species in mm if the penguin weighed 0 grams. It's not a scientifically meaningful value.

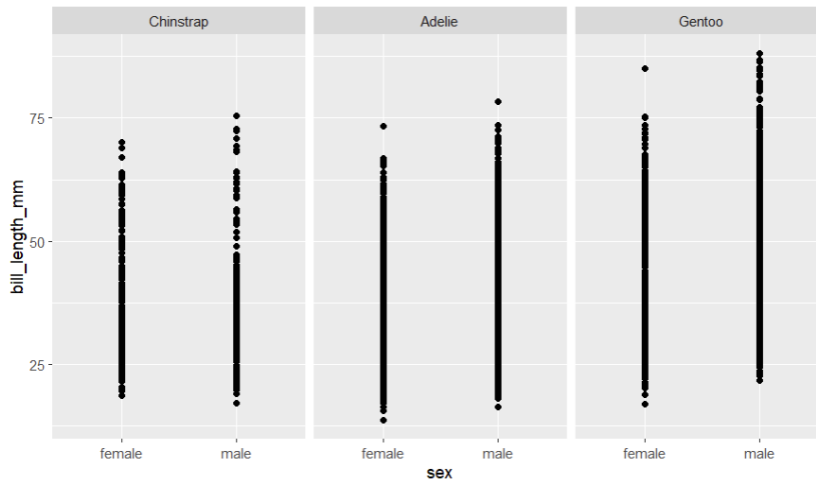
For every increase of a penguin's mass by one gram, their beak length increases by 0.0066mm.

It is better to multiply the coefficients and say that for every increase of a penguin's mass by one kilogram, it's bill length increases by 6.64mm

The residuals look fairly normally distributed, but what might be an issue? Could heteroscedasticity be a problem?

Could try transforming the data, or consider non-parametric regressions that we will not cover here.

Bill Length by Species and Sex



10/6/2026

38

Seems like males might have longer bills in Adelie and Gentoo, and not for Chinstrap, but it is difficult to tell

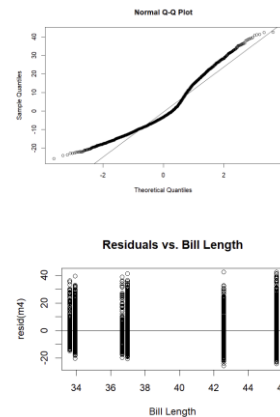
Linear Regression Interpretation 4

```
Call:
lm(formula = bill_length_mm ~ sex + species, data = peng)

Residuals:
    Min       1Q   Median       3Q      Max
-25.770  -8.829  -3.281   7.609  42.571

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  33.6151    0.5324   63.140 < 2e-16 ***
sexmale       3.0593    0.4232    7.230 5.94e-13 ***
speciesAdelie  0.3065    0.5869    0.522  0.602
speciesGentoo  8.9546    0.6079   14.730 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 12.38 on 3426 degrees of freedom
Multiple R-squared:  0.118,    Adjusted R-squared:  0.1172
F-statistic: 152.8 on 3 and 3426 DF,  p-value: < 2.2e-16
```



10/6/2026

39

33.61mm is the bill length of Chinstrap females.

The bill length of males is on average 3.06 mm longer than the bills of females of the same species (this difference is statistically significant).

Bill lengths of Adelie females are 0.307mm longer than bill lengths of Chinstrap females, but this difference isn't significant.

Bill lengths of Gentoo females is 8.95mm longer than female Chinstrap bill lengths, which is statistically significant.

QQplot is not very normal..

Exploring the Palmer Penguin Data

Is it possible that the difference in the bill length between females and males varies by species?

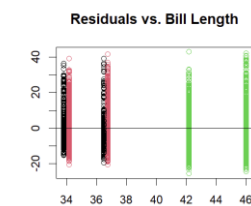
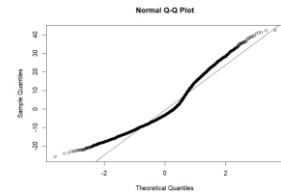
Linear Regression Interpretation 5

```
Call:
lm(formula = bill_length_mm ~ sex * species, data = peng)

Residuals:
    Min       1Q   Median       3Q      Max
-25.363  -8.939  -3.260   7.657  42.837

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)    33.7825     0.6733   50.173 < 2e-16 ***
sexmale         2.6932     0.9955    2.705  0.00686 **
speciesAdelie   0.3751     0.8046    0.466  0.64114
speciesGentoo   8.3807     0.8424    9.949 < 2e-16 ***
sexmale:speciesAdelie -0.1121     1.1764   -0.095  0.92407
sexmale:speciesGentoo  1.1471     1.2180    0.942  0.34635
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 12.38 on 3424 degrees of freedom
Multiple R-squared:  0.1185,    Adjusted R-squared:  0.1172
F-statistic: 92.06 on 5 and 3424 DF,  p-value: < 2.2e-16
```



41

33.78mm – bill length of Chinstrap females. Average length of bills of male Chinstrap is 2.69mm longer than in Chinstrap females.

Bill lengths of female Adelie are 0.375mm longer than bill lengths of female Chinstraps, but this difference is not significant. Bill lengths of female Gentoo are 8.38mm longer than bill lengths of female Chinstrap, which is statistically significant.

-0.1121: difference in bill length between male Adelie and female Adelie is 0.1121mm less than difference in bill length between male Chinstrap and female Chinstrap. But this difference in difference is not significant.

1.147: difference in bill length of male and female Gentoo is 1.147mm bigger, than difference in bill length between male and female Chinstrap, though it's also not significant.

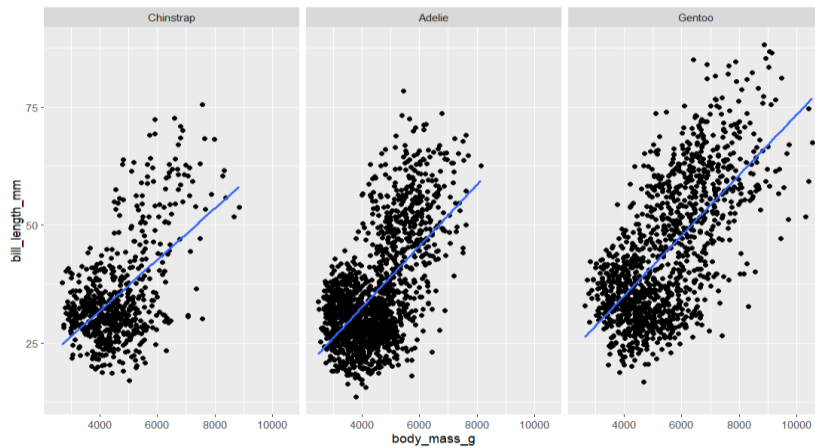
Average bill length of female Adelie: $33.783 + 0.3751 = 34.16$

Average bill length of male Adelie: $33.783 + 2.693 + 0.3751 - 0.1121 = 36.739$

Average bill length of female Gentoo: $33.78 + 8.3807 = 42.161$

Average bill length of male Gentoo: $33.78 + 2.69 + 8.3807 + 1.1471 = 45.99$

Species, Body Mass, and Bill Length



10/6/2026

42

If we look at species, mass, and beak length together, we can see that Gentoo generally might be larger and have a longer beak.

Linear Regression Interpretation 6

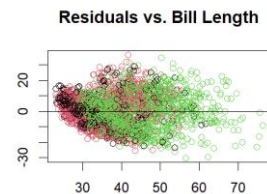
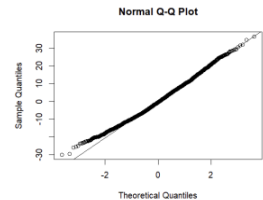
```
Call:
lm(formula = bill_length_mm ~ body_mass_g + species, data = peng)

Residuals:
    Min       1Q   Median       3Q      Max
-30.066  -7.172  -0.502   6.618  36.679

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  6.0538147   0.7391125   8.191 3.63e-16 ***
body_mass_g  0.0062924   0.0001361  46.227 < 2e-16 ***
speciesAdelie 1.4052469   0.4643997   3.026  0.0025 **
speciesGentoo 3.8924911   0.4935059   7.887 4.11e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.788 on 3426 degrees of freedom
Multiple R-squared:  0.4485,    Adjusted R-squared:  0.448
F-statistic: 928.8 on 3 and 3426 DF,  p-value: < 2.2e-16
```

10/6/2026



43

6.054 – bill length of Chinstrap at 0 grams of mass

Bill length of all species of penguins increases by 6.29mm for every kilogram of gained weight, which is statistically significant.

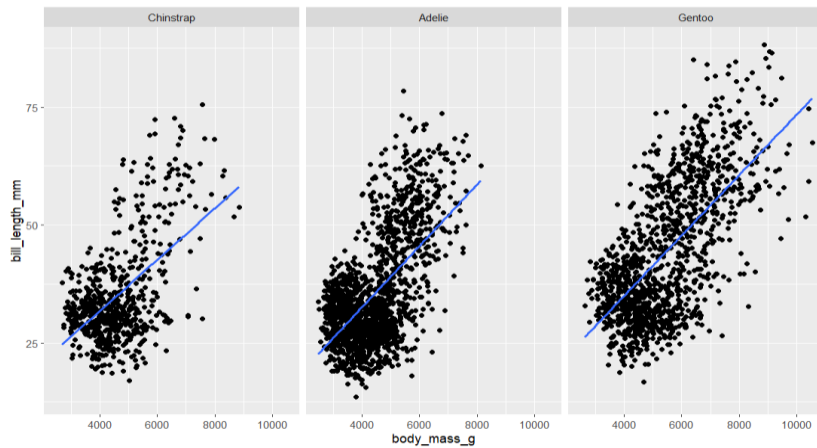
Adelie bill length is 1.405mm longer than Chinstrap bill length when both are at 0 grams of mass, while Gentoo bill length is 3.89mm longer at 0 grams of mass.

Does it make sense to discuss these specific differences? No, since 0 mass penguins do not exist.

This model assumes that the change in bill length relative to body mass for all species is the same. However, this might not be the case!

Residuals QQplot looks normal. The species-specific vertical lines in the residual vs bill length graph are gone since they are masked by the continuous measures of the body mass

Species, Body Mass, and Bill Length



10/6/2026

44

Is there any way we could make this graph a bit easier to read?

Species, Body Mass, and Bill Length Plotted Together



10/6/2026

45

By plotting the graphs over each other, we can see that different species have slightly different slopes in the relationship between body mass and bill length

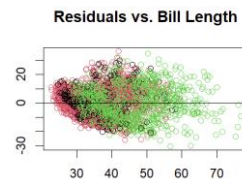
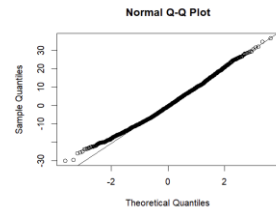
Linear Regression Interpretation 7

```
Call:
lm(formula = bill_length_mm ~ body_mass_g * species, data = peng)

Residuals:
    Min       1Q   Median       3Q      Max
-30.283  -7.176  -0.478   6.561  36.471

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  10.0121114    1.6732204     5.984 2.41e-09 ***
body_mass_g    0.0054323    0.0003534    15.370 < 2e-16 ***
speciesAdelie -3.4802666    1.9832206    -1.755  0.0794 .
speciesGentoo  -0.6645410    1.9867524    -0.334  0.7380 .
body_mass_g:speciesAdelie  0.0010686    0.0004233     2.525  0.0116 *
body_mass_g:speciesGentoo  0.0009701    0.0004014     2.417  0.0157 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.781 on 3424 degrees of freedom
Multiple R-squared:  0.4497,    Adjusted R-squared:  0.4489
F-statistic: 559.5 on 5 and 3424 DF,  p-value: < 2.2e-16
```



46

10.01 – bill lengths of all Chinstrap at 0g mass

Chinstrap bill length increases by 5.4mm for every kilogram it gains, which is statistically significant

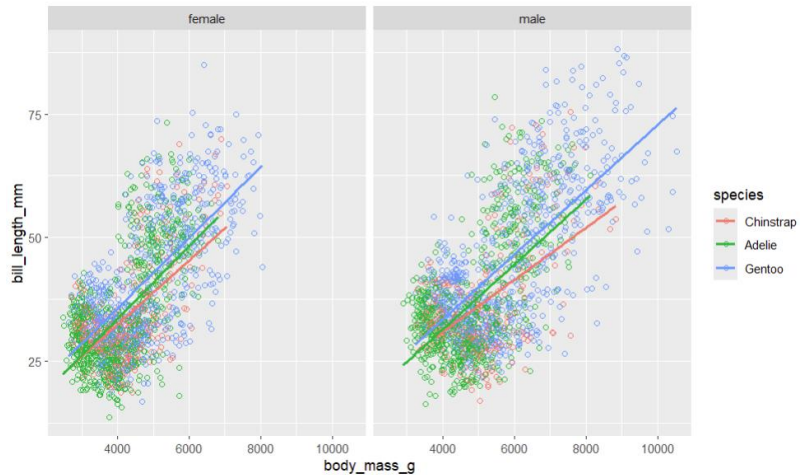
Adelie bill length is 3.48mm shorter than Chinstrap bill length at 0g, while Gentoo bill length is 0.66mm shorter, though neither are significant

For every gained kilogram of mass Adelie bill length increases by 1.07mm more than the bill of Chinstrap for every kilogram it gains (statistically significant)

For every gained kilogram Gentoo bill length increases by 0.970mm more than the bill of Chinstrap for every kilogram it gains (statistically significant)

What will be the beak length of a 6.5kg Gentoo? $10.01 - 0.664 + (0.0054 + 0.0009701) * 6500 = 50.7$

Species, Body Mass, and Bill Length separated by Sex



10/6/2026

47

Interpretations?

Questions?



10/6/2026

48

Interpretations?